

Data Card: SuperBPE Tokenizer for African LLM

Team: Bangweulu Team

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Input vs. Output	Input: Raw text pulled from public multilingual/African-language corpora — news articles, tweets, dialectal Arabic text, religious text, health-domain parallel text, book excerpts, and instruction–response pairs. During preprocessing, this text was converted into ASCII characters then used to train the tokenizer. Output: a learned vocabulary of subword and cross-word merges along with their tokenized sequences (token ids).
Classification problem	Not a classification task. Tokenizer training is unsupervised — it learns merge rules from raw text frequency, not from labeled categories. Some source datasets carry their own labels (news topics in masakhanews; positive/negative sentiment in AfriSenti, Fulah, and Akan corpora), but those labels are not used — only the raw text fields are ingested.
Labeling	No new labels are created for this project. Where the source datasets include labels, they originate with the original authors: journalists (masakhanews), Twitter users plus human annotators (AfriSenti), and domain curators for the sentiment corpora (Fula, Akan). This project does not audit or re-verify those labels since it only uses the underlying text.
Data creation	Primarily aggregated from existing, publicly published Hugging Face datasets (13 sources) rather than collected first-hand — no interviews or original recordings. One gap language, Afrikaans, was filled with synthetic data: English text generated by AI and machine-translated to Afrikaans, since no suitable public corpus was found. Several other under-resourced languages (Wolof, Fula, Kanuri, Zarma, Nuer/Dinka) rely on community-built or machine-translated corpora rather than native speaker-collected text.
Web collection ethics	Most datasets were sourced as already-published resources from the Hugging Face Hub. However, the isiZulu dataset was collected directly from the internet by scraping the webpages provided as source links, which facilitated and streamlined the data collection process. The collection methods for the other datasets vary according to their original sources: AfriSenti, for example, was collected from Twitter, while others were compiled by researchers from books, translations, or community contributions. The one AI-generated dataset (Afrikaans) was produced internally through translation rather than web scraping. Copyright/consent status is incomplete: confirmed licenses so far are CC BY 4.0 (AfriSenti, Kanuri book corpus, Wori-Wolof), CC BY-SA 4.0 (isiZulu news), CC BY-NC 4.0 (Alexandria Dialectal Arabic), CC 4.0 Non-Commercial (masakhanews), MIT (arabic-dialect-corpus, Akan sentiments, Fulah sentiment corpus), and Apache-2.0 (Chichewa machine translation). Four datasets (health translation, bible zarma, Bambara raw corpus, Afrikaans) still have no confirmed license — this must be resolved before any redistribution of the trained tokenizer or its training data.
Bias and representation	Based on the current dataset composition: coverage skews toward languages that already have Hugging Face/NLP research activity — via the two multilingual sets

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	(masakhanews, AfriSenti: several languages), plus single-language sets for Zulu, Kanuri, Fula, Chichewa, Zarma, Bambara, Akan, Wolof, 6 Arabic dialects, and Afrikaans. That's roughly two dozens-plus languages out of the 2,000+ spoken across Africa, so most language families are simply absent from this corpus. Representation is also uneven in depth: several languages (Fula, Akan, Wolof) are represented by only one narrow-domain dataset (sentiment-only or instruction-only text), and Afrikaans is represented purely by synthetic, machine-translated text rather than native writing — both known risks for skewing what the tokenizer learns to merge.
Values and community alignment	The corpus was assembled to build an open, general-use tokenizer for under-served African languages rather than to optimize a single high-resource language — prioritizing breadth of language coverage within a modest sized vocabulary. Community benefit: an open, validated tokenizer lowers the barrier for future African-language model builders. Accountability is maintained by publishing this data card alongside per-dataset source links, licenses, and citations, so downstream users can trace provenance and respect each original dataset's terms.

Dataset Sources & Citations

Dataset	Language(s)	License	Citation
Masakhanews	Amharic (amh), Hausa (hau), Igbo (ibo), Lingala (lin), Luganda (lug), Oromo (orm), Nigerian Pidgin (pcm), Rundi (run), chShona (sna), Somali (some), Kiswahili (swa), Tigrinya (tir), isiXhosa (xho), Yoruba (yor)	CC 4.0 Non-Commercial	Adelani et al., 2023, “MasakhaNEWS: News Topic Classification for African Languages,” ArXiv
AfriSenti	Algeria Arabic, Amharic, Hausaz Igbo, Kinyarwanda, Moroccan Arabic/Darija, Oromo, Swahili, Xithonga, Yoruba	CC BY 4.0	Muhammad et al., 2023, “AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages”
Arabic-dialect-corpus	Masri (Egyptian), Maghrebi (North Africa)	MIT	Dataflare, 2026, “Arabic Dialect Corpus,” Hugging Face
isiZulu news	Zulu	CC BY-SA 4.0	Madodonga, Marivate & Adendorff, 2023, “Izindaba-Tindzaba: Machine learning news categorisation for Long and Short Text for isiZulu and Siswati,” DOI: 10.55492/dhasa.v4i01.4449

Dataset	Language(s)	License	Citation
Kanuri book corpus	Kanuri	CC BY 4.0	Öktem, Jaam, DeLuca & Tang, “Gamayun – Language Technology for Humanitarian Response,” IEEE GHTC 2020
Fulah sentiment corpus	Fula	MIT	Owusu, 2025, “Fulah Sentiment Corpus”
Health translation dataset	Nuer, Dinka	--	dayomtechnologies, 2026, “English-Nuer-Dinka-Health-Translation-Dataset,” Hugging Face (machine-translated from DigitalUmuganda/NMT_Health_parallel_data_en_kin)
Chichewa machine translation	Chichewa	Apache-2.0	Matekenya, D., 2024, “chichewa-machine-translation,” Hugging Face
Bible zarma	Zarma	--	Abdou Aziz, “bible_zarma,” Hugging Face (no formal publication found)
Bambara texts raw corpus	Bambara	--	Makan09, “Bambara_texts_raws_corpus,” Hugging Face (no formal publication found)
Akan sentiments corpus	Akan	MIT	Owusu, 2025, “Akan Sentiment Corpus”
Wori-wolof instruction	Wolof	CC BY 4.0	Diop, 2026, “WORl: Wolof Reverse Instruction Dataset,” Hugging Face
Alexandria Dialectal Arabic	Algerian Arabic, Egyptian Arabic, Libyan Arabic, Mauritanian Arabic, Morocco Arabic, and Sudan Arabic	CC BY-NC 4.0	EL Mekki et al., 2026, “Alexandria: A Multi-Domain Dialectal Arabic MT Dataset,” ACL 2026
Afrikaans	Afrikaans	--	AI-generated (synthetic English → Afrikaans translation), no external source